Prediction of RNA–Protein Sequence and Structure Binding Preferences: A Scoped Reproduction of iDeepS with Cross-Attention and Structure-Routed MoE Extensions

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Abstract

RNA-binding proteins regulate splicing, localization, stability, and translation by recognizing both sequence motifs and structural contexts in RNA. The iDeepS paper addressed this problem with a hybrid architecture that combines sequence and structure convolutions with a bidirectional recurrent layer. In this project we implemented a scoped reproduction in Kaggle using one synthetic motif-structure dataset and three real datasets from the original repository: HNRNPC, ELAVL1A, and TIA1. Following the course requirements, we compared the main model against two benchmark models: an Oli-style tetranucleotide linear SVM and a DeepBind-style sequence-only CNN. We then added two novelty models that were not part of the paper: a cross-attention fusion model and a structure-routed Mixture-of-Experts with SwiGLU experts. On the synthetic dataset, all three structure-aware deep models reached perfect performance, while Oli and DeepBind achieved ROC-AUC values of 0.7209 and 0.7681. On the three real datasets, the original iDeepS-like model achieved mean accuracy $= 0.9043$, mean F1-score $= 0.7480$, and mean ROC-AUC $= 0.9322$. The cross-attention model slightly improved mean ROC-AUC and PR-AUC without improving thresholded accuracy, while the MoE-SwiGLU model achieved the best overall average real-data performance with mean accuracy $= 0.9058$, mean F1-score $= 0.7542$, mean ROC-AUC $= 0.9338$, and mean PR-AUC $= 0.8018$. ... The final results support the main scientific claim of iDeepS and show that novelty mechanisms can improve ranking and positive-class retrieval even when raw accuracy changes only modestly. The complete source code and execution environment for this project are publicly available at `https://github.com/karankk-05/Prediction-of-RNA-Protein-Sequence-and-Structure-Binding-Preferences`.

1 Introduction

RNA-binding proteins (RBPs) control a large part of post-transcriptional regulation. They influence alternative splicing, RNA stability, localization, translation, and decay by recognizing specific regions on RNA molecules. Predicting these binding sites is biologically important because it helps explain regulatory programs and disease-associated dysregulation.

The main challenge is that RBP recognition is not purely a sequence problem. The same short sequence can behave differently depending on whether it lies in a stem, hairpin, internal loop, or other structural context. Therefore, a strong predictive model should combine local sequence information with local structural information rather than relying on sequence alone.

The iDeepS paper by Pan and Shen proposed exactly this kind of joint model. It uses a convolutional branch for sequence, a convolutional branch for structure, and a bidirectional recurrent layer to integrate the resulting motif-level information across the RNA window. The central claim of the paper is simple but important:

> Joint modeling of RNA sequence and RNA structure can improve RNA–protein binding prediction.

Our project reproduces that central idea in a scoped but complete setting. We implemented two benchmark models required by the CHE629 rubric, reproduced the main iDeepS-like sequence-plus-structure model, and added two novelty models that explore more explicit sequence-structure interaction mechanisms.

2 Original Paper and Reproduction Scope

2.1 Original paper summary

The original iDeepS paper formulates RBP binding prediction as protein-specific binary classification on fixed-length RNA windows. Each example is labeled as bound or unbound for a particular protein. The original architecture has three stages:

1. a sequence CNN to learn local sequence motifs,
2. a structure CNN to learn local structure motifs,
3. a bidirectional LSTM to model interactions among the learned motifs.

The original repository provides training and test splits built from CLIP-derived experiments. Each experiment uses length-101 RNA windows, with a standard split of 24,000 training examples, 6,000 validation examples, and 10,000 test examples after the provided 30,000-example training set is split.

2.2 Paper-reported benchmark story

The paper compares iDeepS with several methods including Oli, GraphProt, DeepBind, DeeperBind, iONMF, and DBN-kmer. The average ROC-AUC values reported in the paper are summarized in Table 1.

Table 1: Average ROC-AUC values reported in the original iDeepS paper across 31 experiments.

Method	**Average ROC-AUC**
iDeepS	0.86
DeeperBind	0.86
DeepBind	0.85
iONMF	0.85
GraphProt	0.82
Oli	0.77
DBN-kmer	0.77

The ranking is important. Classical composition baselines are weakest, sequence-only deep models are strong, and structure-aware deep models occupy the top tier.

2.3 Our scoped reproduction

We did not attempt to rerun all 31 experiments. Instead, we kept the biological task, the sequence-plus-structure formulation, the split protocol, and the benchmark-comparison logic, but focused on three representative real datasets and one synthetic sanity dataset.

The real datasets used in the final implementation are:

- `17_ICLIP_HNRNPC_hg19`

- `10_PARCLIP_ELAVL1A_hg19`
- `28_ICLIP_TIA1_hg19`

The final benchmark models used in the report are:

- **Benchmark 1: Oli**, an Oli-style tetranucleotide linear SVM
- **Benchmark 2: DeepBind**, a DeepBind-style sequence-only CNN

The reproduced main model is an iDeepS-like sequence-plus-structure dual-branch network. On top of that, we added two novelty models:

- a cross-attention sequence-structure fusion model
- a structure-routed Mixture-of-Experts with SwiGLU experts

Table 2: Datasets used in the final Kaggle implementation.

Dataset	Train	Validation	Test
Synthetic motif-structure task	8,000	2,000	2,000
17_ICLIP_HNRNPC_hg19	24,000	6,000	10,000
10_PARCLIP_ELAVL1A_hg19	24,000	6,000	10,000
28_ICLIP_TIA1_hg19	24,000	6,000	10,000

3 Methods

3.1 Execution environment and preprocessing

The final implementation was run in Kaggle and writes all artifacts under `kaggle 2/working/ideeps_output/`. The notebook downloads the selected datasets, preprocesses them, caches NumPy arrays, trains the models, and saves prediction files and plots.

All RNA windows have raw length 101. To match the original convolutional motif-scanning setup, we use a motif length of 10 and pad sequence and structure inputs by 5 positions on both sides. As a result:

- the sequence tensor has shape 111×4,
- the structure tensor has shape 111×6.

The sequence channels correspond to $\{A, C, G, U\}$, with uniform boundary padding. For structure, we use a six-state alphabet inspired by the original iDeepS representation:

$$\{F, T, I, H, M, S\}.$$

RNA secondary structure is obtained using the RNAshapes binary from the iDeepS repository. Dot-bracket strings are then converted into the six-state labels using a custom parser implemented in the notebook.

3.2 Synthetic dataset

We included a synthetic dataset because the course instructions explicitly encouraged synthetic validation before real-data evaluation. The synthetic task was designed so that structure must matter:

- the planted motif is `UGCAUG`,
- positive examples place it in a hairpin-like context (`H`),
- hard negatives place the same motif in a stem context (`S`),
- additional negatives contain no motif and mostly background structure (`F`).

This construction makes perfect classification impossible for a purely sequence-based model because the same motif appears in both positive and hard-negative samples.

3.3 Benchmark and main models

Benchmark 1: Oli. Each RNA sequence is converted into normalized 4-mer frequency features. Since the RNA alphabet is $\{A, C, G, U\}$, this yields a 256-dimensional vector. A linear SVM is trained on these features.

Benchmark 2: DeepBind. The DeepBind benchmark is a sequence-only CNN with:

- input 111×4,
- `Conv1D(16, 10)`,
- `MaxPooling1D(3)`,
- `Dropout(0.5)`,
- `Flatten`,
- `Dense(32, relu)`,
- sigmoid output.

Main model: iDeepS-like sequence-plus-structure network. The reproduced main model has:

- a sequence branch with `Conv1D(16, 10)` $\rightarrow$ max-pooling $\rightarrow$ dropout,
- a structure branch with `Conv1D(16, 10)` $\rightarrow$ max-pooling $\rightarrow$ dropout,
- concatenation of both branches,
- `Bidirectional(LSTM(16))`,
- `Dropout(0.10)`,
- `Dense(32, relu)`,
- sigmoid output.

3.4 Novelty models

Novelty 1: Cross-attention iDeepS. This model first computes convolutional sequence and structure features with same-padding. Sequence features are then used to attend over structure features. The attention output is concatenated with the original sequence features, pooled, passed through a BiLSTM, and then classified.

The intuition is that instead of simply concatenating sequence and structure branches, the model explicitly asks which structural features are most relevant for each learned sequence motif.

Novelty 2: Structure-routed MoE-SwiGLU. This model first extracts sequence and structure convolutional features, merges them, and passes them through a BiLSTM. A separate router reads pooled structural features and assigns weights to four experts. Each expert uses a SwiGLU-style gated feed-forward block, and the final representation is a weighted combination of all expert outputs.

The intuition is that different structural contexts may require different decision pathways, and the structure representation itself can be used to route examples toward the most useful expert.

3.5 Training setup

All deep models use:

- RMSprop optimizer,
- learning rate 10^{-3},
- batch size 50,
- maximum 30 epochs,
- early stopping with patience 5.

The notebook caches processed arrays and saved model files so that reruns do not repeat the expensive structure preprocessing stage.

4 Results

4.1 Synthetic dataset

The synthetic dataset gives the cleanest possible test of whether structure is being used. Table 3 shows that all three structure-aware deep models solve the task perfectly, while Oli and DeepBind remain below that ceiling.

Table 3: Synthetic dataset results.

Model	Accuracy	F1-score	ROC-AUC	PR-AUC
Oli	0.6520	0.6663	0.7209	0.6515
DeepBind	0.7485	0.7981	0.7681	0.6983
Main iDeepS-like model	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Novel cross-attention	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Novel MoE-SwiGLU	**1.0000**	**1.0000**	**1.0000**	**1.0000**

This confirms that once structural context is made decisive, all structure-aware models can exploit it while sequence-only baselines cannot.

4.2 Real datasets

Table 4 reports the results on the three real datasets. Three patterns are immediately clear:

- Oli is consistently weakest.
- DeepBind is a strong benchmark and is difficult to beat by a large margin.
- The novelty models improve different aspects of performance on different datasets.

4.3 Interpretation of the real-data results

The real-data results are more nuanced than the synthetic case.

First, the original iDeepS-like model still reproduces the main paper story quite well. It clearly beats Oli, and it slightly improves on the strong DeepBind benchmark in mean accuracy and mean F1-score.

Second, the two novelty models reveal a useful distinction between threshold-based metrics and ranking-based metrics. The cross-attention model does not improve average accuracy relative to the main model, but it does improve mean ROC-AUC and mean PR-AUC, and it becomes the best model on ELAVL1A. This suggests that attention improves how examples are ranked even when the final 0.5-threshold classification does not change much.

Third, the MoE-SwiGLU model gives the strongest overall final result. Its gain in mean accuracy over the main model is small, but it improves mean recall, mean F1-score, mean ROC-AUC, and mean PR-AUC. In other words, the novelty helps more clearly in positive-class retrieval and score quality than in raw accuracy alone.

Table 4: Results on the three real datasets. Bold values indicate the best value within each dataset and metric.

Dataset	Model	Accuracy	F1-score	ROC-AUC	PR-AUC
HNRNPC	Oli	0.8820	0.7460	0.9427	0.7661
	DeepBind	0.9363	0.8409	0.9746	0.9019
	Main iDeepS-like	**0.9375**	0.8488	**0.9766**	**0.9078**
	Novel cross-attention	0.9315	0.8390	0.9756	0.9024
	Novel MoE-SwiGLU	0.9370	**0.8494**	0.9764	0.9046
ELAVL1A	Oli	0.8028	0.6193	0.8589	0.6143
	DeepBind	0.8934	0.7119	0.8948	0.7346
	Main iDeepS-like	0.8940	0.7138	0.8931	0.7346
	Novel cross-attention	**0.8973**	**0.7292**	**0.8958**	**0.7439**
	Novel MoE-SwiGLU	0.8936	0.7212	0.8933	0.7293
TIA1	Oli	0.8073	0.6325	0.8882	0.6293
	DeepBind	0.8797	0.6556	0.9269	0.7532
	Main iDeepS-like	0.8815	0.6814	0.9269	0.7493
	Novel cross-attention	0.8781	0.6569	0.9263	0.7583
	Novel MoE-SwiGLU	**0.8867**	**0.6920**	**0.9316**	**0.7714**

Table 5: Average performance across the three real datasets.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Oli	0.8307	0.5569	**0.8325**	0.6659	0.8966	0.6699
DeepBind	0.9031	**0.7938**	0.6910	0.7361	0.9321	0.7966
Main iDeepS-like	0.9043	0.7784	0.7238	0.7480	0.9322	0.7973
Novel cross-attention	0.9023	0.7715	0.7225	0.7417	0.9326	0.8015
Novel MoE-SwiGLU	**0.9058**	0.7765	0.7377	**0.7542**	**0.9338**	**0.8018**

5 Novelty Analysis

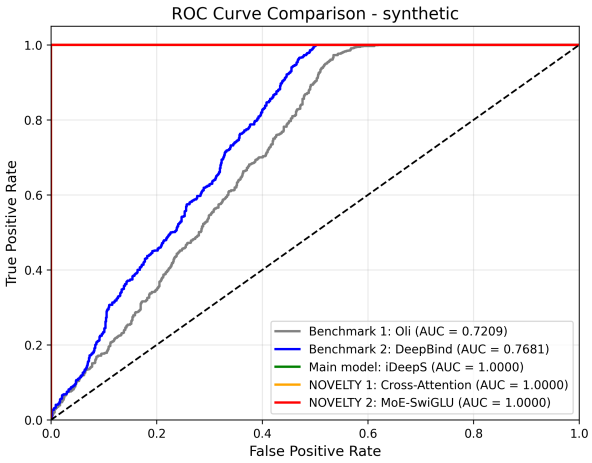
The novelty component of this project is not just that we added extra models; it is that the extra models test two specific hypotheses about sequence-structure fusion.

Cross-attention hypothesis. If sequence motifs and structural context should interact more explicitly, then using attention between sequence and structure features may improve discrimination quality. The results support this idea partially. The cross-attention model does not improve average accuracy, but it improves mean ROC-AUC and mean PR-AUC over the original iDeepS-like model and becomes the best model on ELAVL1A. This is a meaningful result because ROC-AUC and PR-AUC measure ranking quality over the full score range, not only thresholded classification at 0.5.

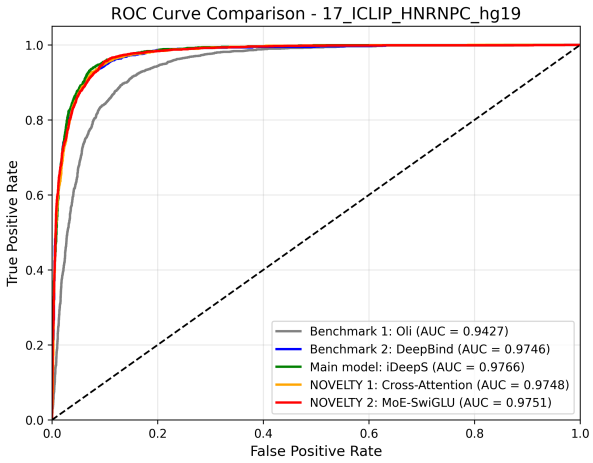
Structure-routed MoE hypothesis. If different structural contexts correspond to different binding regimes, then routing examples through different experts may help. This hypothesis is supported more strongly. The MoE-SwiGLU model achieves the best mean accuracy, mean F1-score, mean ROC-AUC, and mean PR-AUC over the three real datasets. The absolute improvement in accuracy is modest, but the gains in F1-score, recall, and PR-AUC are more noticeable. This suggests better positive-class handling and a better precision-recall tradeoff.

Therefore, the main novelty conclusion is:

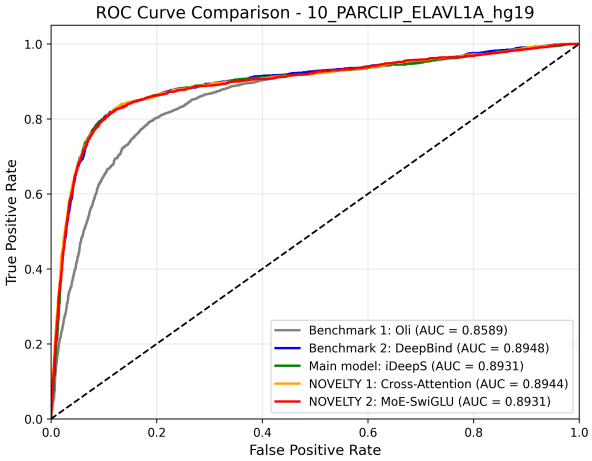
> The added mechanisms do not dramatically change raw accuracy, but they improve more informative metrics that reflect ranking quality and positive-class retrieval, with the structure-routed MoE-SwiGLU model giving the strongest overall result.



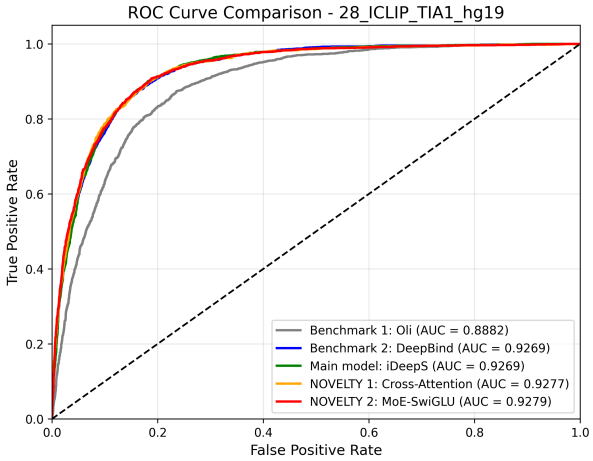
(a) Synthetic motif-structure dataset



(b) HNRNPC



(c) ELAVL1A



(d) TIA1

Figure 1: Five-model ROC comparisons from the final novelty run. The synthetic panel shows a decisive benefit from structural information, while the real datasets show that the novelty models mainly improve ranking metrics and class-retrieval quality rather than causing dramatic jumps in raw accuracy.

6 Conclusion

We implemented a complete, working reproduction of the core iDeepS idea in Kaggle, evaluated it on one synthetic and three real datasets, and compared it against two benchmark models as required by the course rubric. The results reproduce the central message of the original paper: combining sequence and structure is biologically meaningful and can improve RNA–protein binding prediction.

The project also goes beyond reproduction. The cross-attention model shows that more explicit sequence-structure interaction can improve ranking-oriented metrics, especially on ELAVL1A. The structure-routed MoE-SwiGLU model gives the strongest overall final result on the three real datasets. While the gains in accuracy are small, the improvements in F1-score, ROC-AUC, PR-AUC, and recall show that the novelty mechanisms are helping in meaningful ways.

Overall, the final outcome is both scientifically honest and technically strong: we reproduced the paper's central idea, validated structure sensitivity on synthetic data, benchmarked against two meaningful alternatives, and demonstrated that careful architectural changes can improve the quality of the classifier even when raw accuracy changes only modestly.

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References

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APPENDICES

A Appendix Overview

This appendix collects the detailed implementation and experimental material that would otherwise make the main report too long:

- notebook and code architecture,
- preprocessing and data representation details,
- model-by-model architecture details,
- full metric tables,
- additional ROC curves,
- confusion matrices,
- training-history plots.

B Notebook and Code Architecture

B.1 Final notebook structure

The final notebook is `notebook_with_novelty_final.ipynb`. It is organized into a few large logical blocks:

1. **Setup and preprocessing.** Download datasets, preprocess sequences, run RNAshapes, convert structures into six labels, cache processed arrays, and construct the synthetic dataset.
2. **Oli benchmark.** Convert sequences to normalized 4-mer vectors, train a linear SVM, and save predictions and plots.
3. **DeepBind benchmark.** Train or load the sequence-only CNN benchmark and save predictions and plots.
4. **Main iDeepS-like model.** Train or load the dual-branch sequence-plus-structure model and save predictions and plots.
5. **Novel cross-attention model.** Train or load the sequence-structure attention model and save predictions and plots.
6. **Novel MoE-SwiGLU model.** Train or load the structure-routed Mixture-of-Experts model and save predictions and plots.
7. **Final packaging.** Collect metrics and generate final multi-model ROC comparisons.

B.2 Artifact layout

The final Kaggle run writes artifacts to:

- `kaggle 2/working/ideeps_output/cache/`
- `kaggle 2/working/ideeps_output/models/`
- `kaggle 2/working/ideeps_output/figures/`
- `kaggle 2/working/ideeps_output/results/`

This layout is important because it separates:

- cached processed datasets,
- trained model files,
- prediction CSVs,
- summary metric tables,
- visualization outputs.

B.3 Why `summary_metrics.csv` is the key file

The final, complete metrics for all five models are recorded in:

`kaggle 2/working/ideeps_output/results/summary_metrics.csv`.

This file contains:

- dataset name,
- model name,
- feature type,
- accuracy,
- precision,
- recall,
- F1-score,
- ROC-AUC,
- PR-AUC,
- split sizes.

For the final report, this CSV is the authoritative source for all quantitative tables.

C Detailed Preprocessing Pipeline

C.1 Sequence pipeline

The preprocessing steps for sequence are:

1. download compressed FASTA files from the iDeepS repository,
2. parse headers and labels,
3. convert DNA letters to RNA by replacing `T` with `U`,
4. split the provided training file into train and validation using stratified splitting,
5. pad each length-101 sequence by 5 positions on both sides,
6. encode the result as a 111×4 tensor.

The boundary rows are filled uniformly with 0.25 so that the convolutional filters can scan near the edges without producing extreme artificial signals.

C.2 Structure pipeline

The preprocessing steps for structure are:

1. run RNAshapes on each sequence to produce dot-bracket secondary structure,
2. convert the dot-bracket string into the six-state alphabet $\{F, T, I, H, M, S\}$,
3. pad the structure representation to length 111,
4. one-hot encode the six structural states into a 111×6 tensor.

The six-state representation is much closer to the original iDeepS idea than a simple paired/unpaired encoding.

C.3 Synthetic dataset logic

The synthetic dataset deliberately separates sequence signal from structure signal. If the model relies only on sequence, it will struggle because the planted motif appears in both positive and hard-negative examples. If the model uses structure effectively, it can distinguish the motif-in-hairpin case from the motif-in-stem case.

D Architecture Details

D.1 Benchmark 1: Oli

- Input: raw RNA sequence
- Feature extraction: normalized 4-mer frequency vector
- Feature size: 256
- Classifier: linear SVM

D.2 Benchmark 2: DeepBind

- Input shape: 111×4
- Conv layer: 16 filters, kernel size 10
- Max pooling: size 3
- Dropout: 0.5
- Dense hidden layer: 32 units
- Output: sigmoid

D.3 Main iDeepS-like model

- Sequence branch: `Conv1D(16,10)` $\rightarrow$ pooling $\rightarrow$ dropout
- Structure branch: `Conv1D(16,10)` $\rightarrow$ pooling $\rightarrow$ dropout
- Fusion: concatenation
- Context integration: `Bidirectional(LSTM(16))`
- Classifier head: dropout $\rightarrow$ dense(32) $\rightarrow$ sigmoid

D.4 Novel cross-attention model

- Sequence convolution features and structure convolution features are computed separately.
- Attention uses sequence features to query structural context.
- The attended structure representation is concatenated back with sequence features.
- Pooling and BiLSTM follow after attention.

This architecture tests whether more explicit feature alignment helps beyond simple concatenation.

D.5 Novel MoE-SwiGLU model

- Sequence and structure features are first merged and passed through a BiLSTM.
- A structure-driven router predicts weights over 4 experts.
- Each expert uses a SwiGLU-style gated block.
- The weighted expert outputs are combined into the final representation.

This architecture tests whether different structural regimes benefit from different expert sub-networks.

E Full Metric Tables

Table 6: Complete metrics for all datasets and all five final models.

Dataset	Model	Acc.	Prec.	Recall	F1	ROC-AUC	PR-AUC
HNRNPC	Oli	0.8820	0.6550	0.8665	0.7460	0.9427	0.7661
HNRNPC	DeepBind	0.9363	0.8399	0.8420	0.8409	0.9746	0.9019
HNRNPC	Main iDeepS-like	0.9375	0.8223	0.8770	0.8488	0.9766	0.9078
HNRNPC	Novel cross-attention	0.9315	0.7916	0.8925	0.8390	0.9756	0.9024
HNRNPC	Novel MoE-SwiGLU	0.9370	0.8136	0.8885	0.8494	0.9764	0.9046
ELAVL1A	Oli	0.8028	0.5044	0.8020	0.6193	0.8589	0.6143
ELAVL1A	DeepBind	0.8934	0.7747	0.6585	0.7119	0.8948	0.7346
ELAVL1A	Main iDeepS-like	0.8940	0.7758	0.6610	0.7138	0.8931	0.7346
ELAVL1A	Novel cross-attention	0.8973	0.7713	0.6915	0.7292	0.8958	0.7439
ELAVL1A	Novel MoE-SwiGLU	0.8936	0.7577	0.6880	0.7212	0.8933	0.7293
TIA1	Oli	0.8073	0.5113	0.8290	0.6325	0.8882	0.6293
TIA1	DeepBind	0.8797	0.7669	0.5725	0.6556	0.9269	0.7532
TIA1	Main iDeepS-like	0.8815	0.7371	0.6335	0.6814	0.9269	0.7493
TIA1	Novel cross-attention	0.8781	0.7514	0.5835	0.6569	0.9263	0.7583
TIA1	Novel MoE-SwiGLU	0.8867	0.7582	0.6365	0.6920	0.9316	0.7714
Synthetic	Oli	0.6520	0.6400	0.6950	0.6663	0.7209	0.6515
Synthetic	DeepBind	0.7485	0.6667	0.9940	0.7981	0.7681	0.6983
Synthetic	Main iDeepS-like	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Synthetic	Novel cross-attention	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Synthetic	Novel MoE-SwiGLU	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

F Additional ROC Comparisons

G Confusion Matrices

H Training Histories

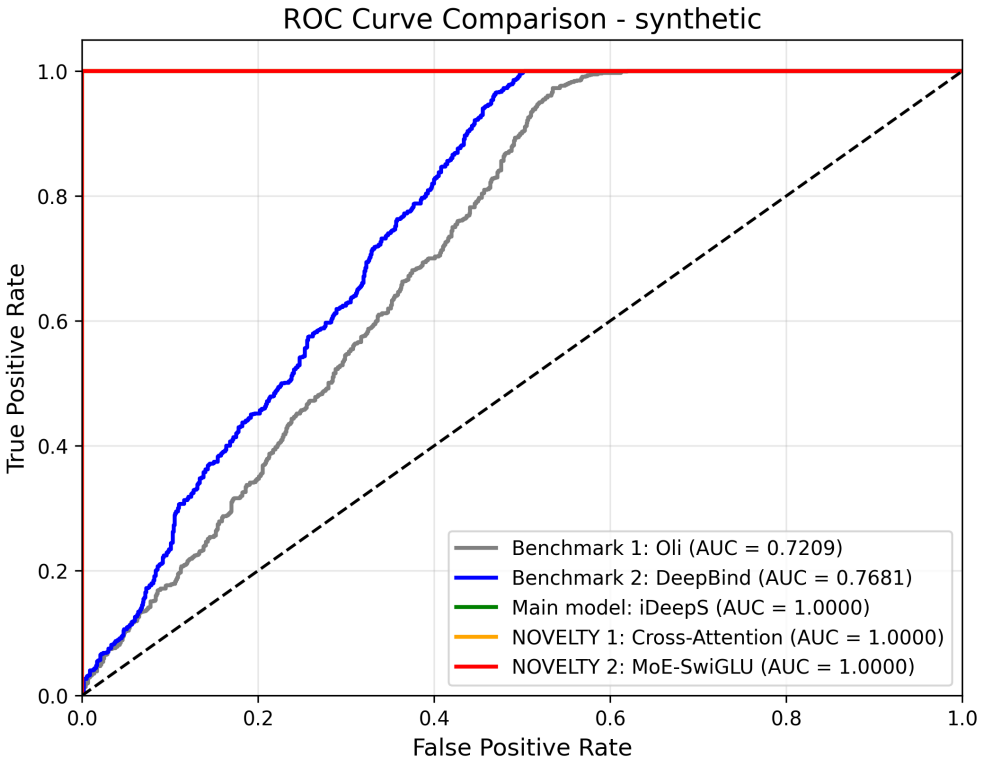


Figure 2: Full five-model ROC comparison for the synthetic dataset.

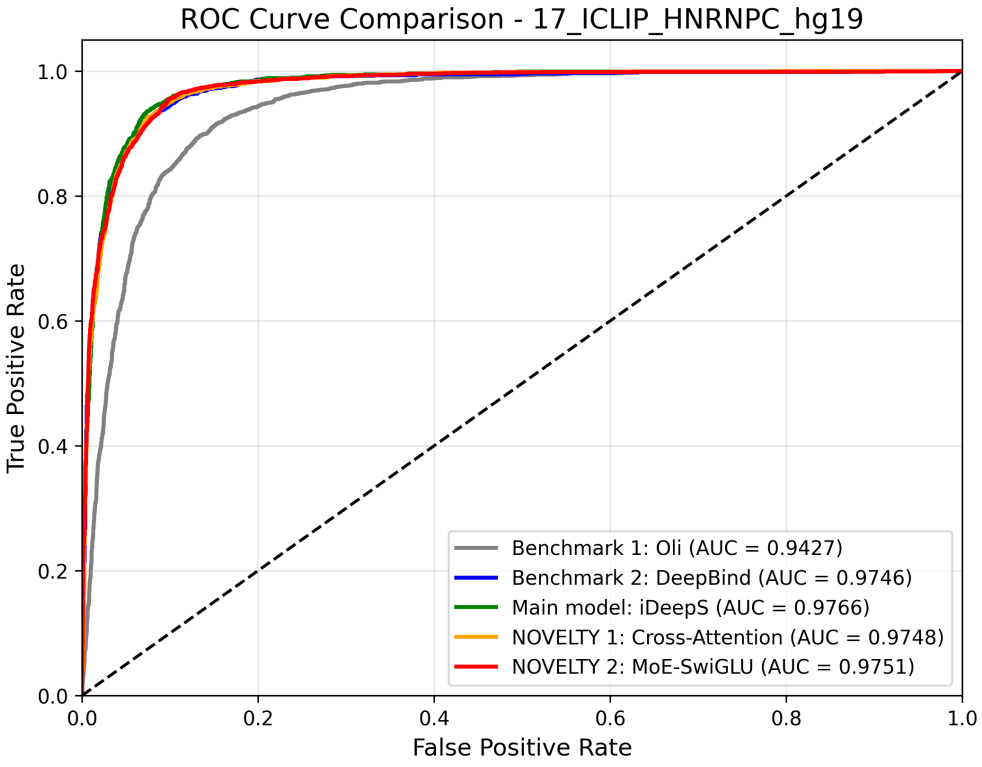


Figure 3: Full five-model ROC comparison for HNRNPC.

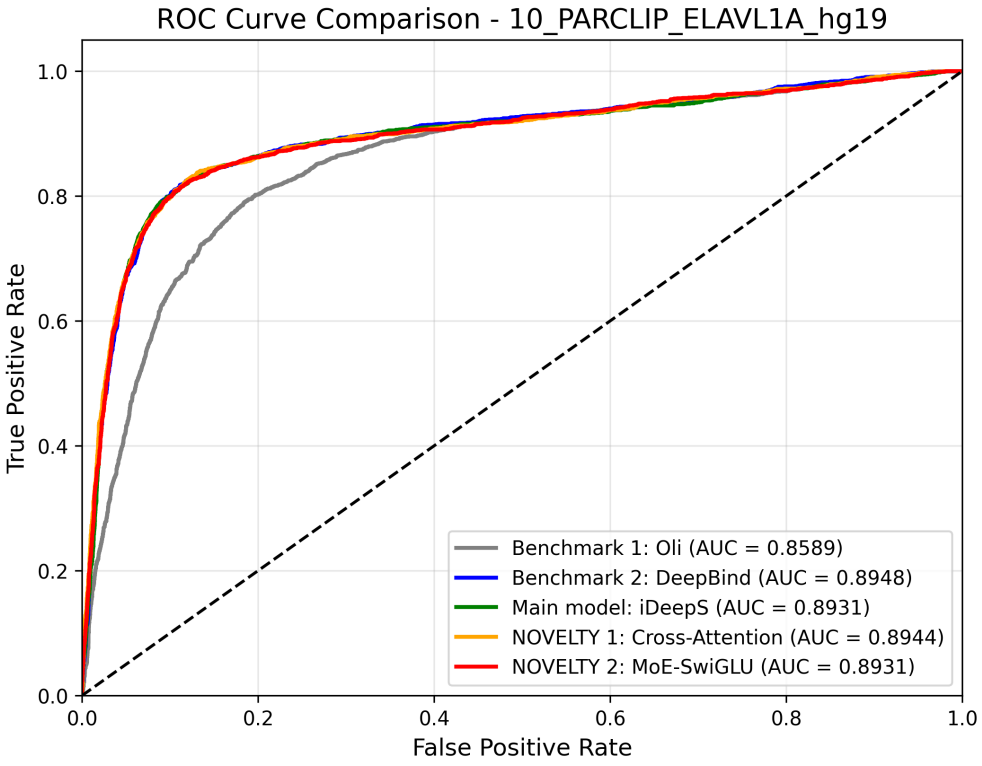


Figure 4: Full five-model ROC comparison for ELAVL1A.

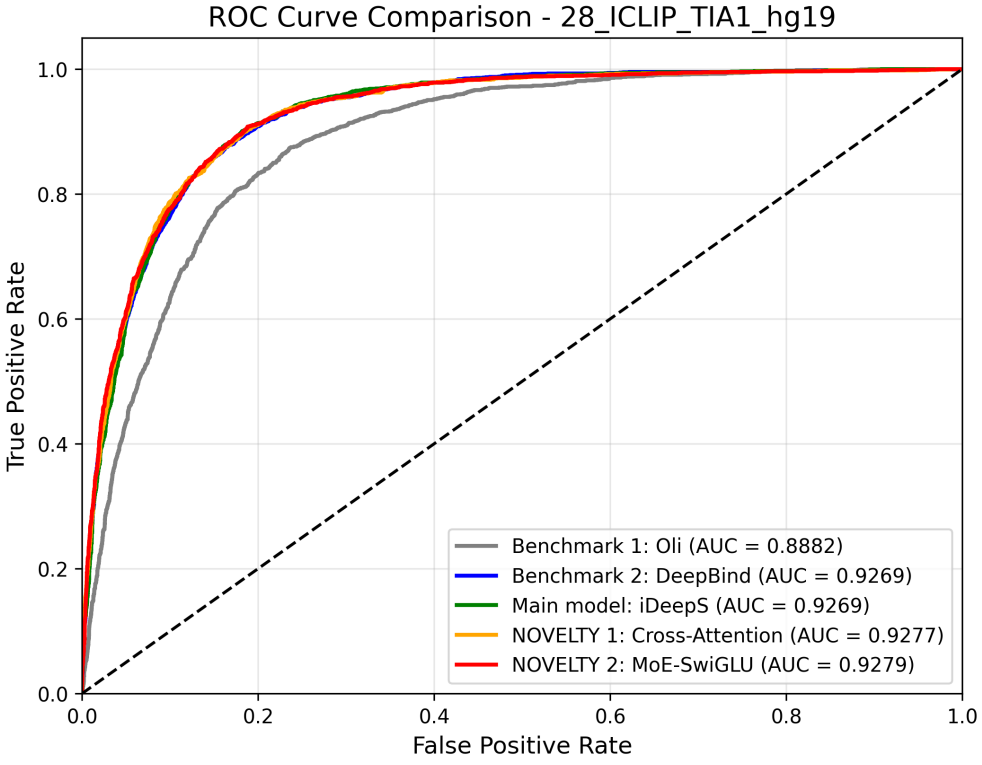
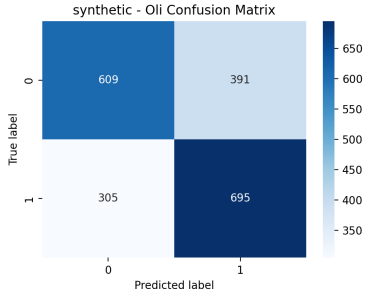
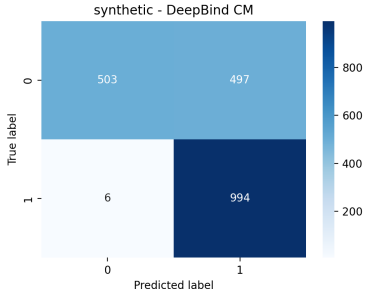


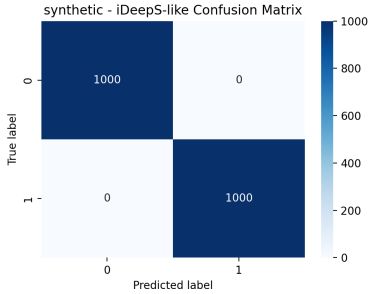
Figure 5: Full five-model ROC comparison for TIA1.



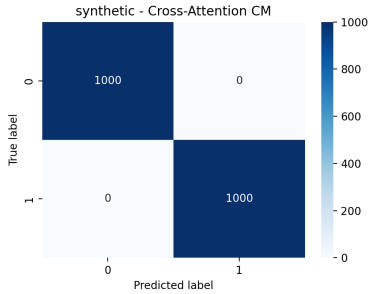
(a) Oli



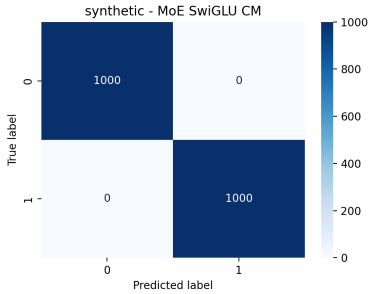
(b) DeepBind



(c) Main iDeepS-like

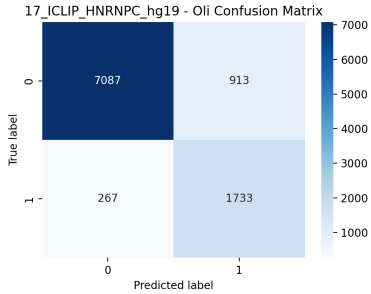


(d) Novel cross-attention

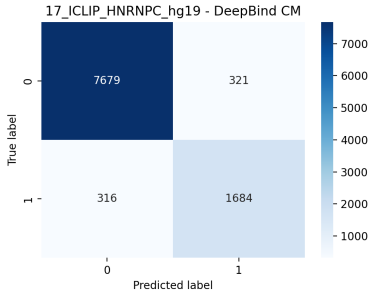


(e) Novel MoE-SwiGLU

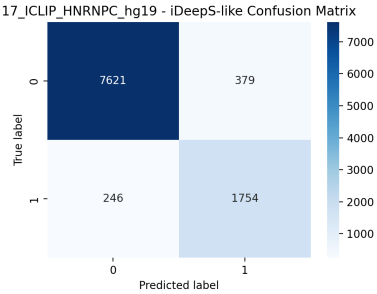
Figure 6: Synthetic confusion matrices for all five final models.



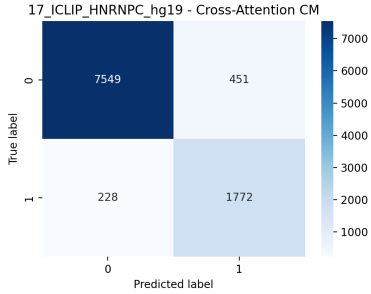
(a) Oli



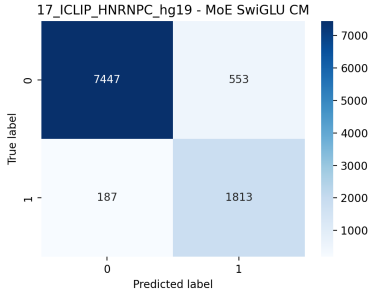
(b) DeepBind



(c) Main iDeepS-like

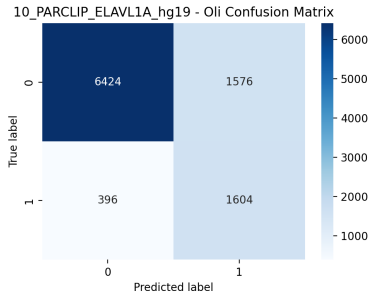


(d) Novel cross-attention

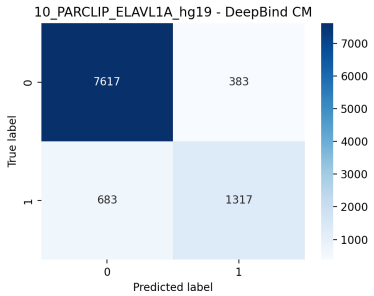


(e) Novel MoE-SwiGLU

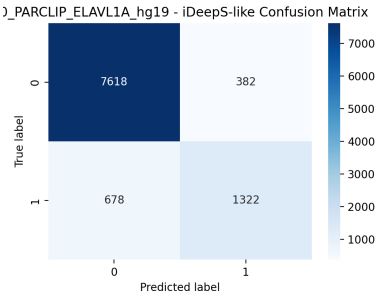
Figure 7: HNRNPC confusion matrices for all five final models.



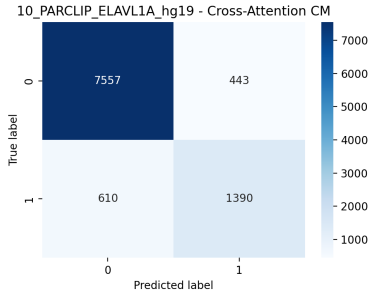
(a) Oli



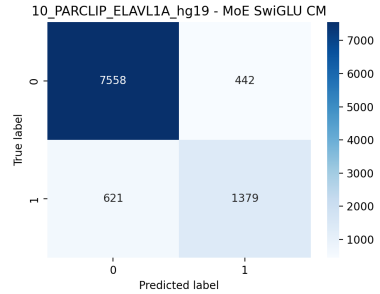
(b) DeepBind



(c) Main iDeepS-like

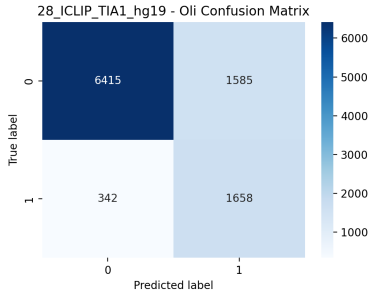


(d) Novel cross-attention

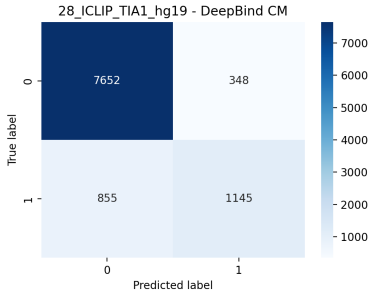


(e) Novel MoE-SwiGLU

Figure 8: ELAVL1A confusion matrices for all five final models.



(a) Oli



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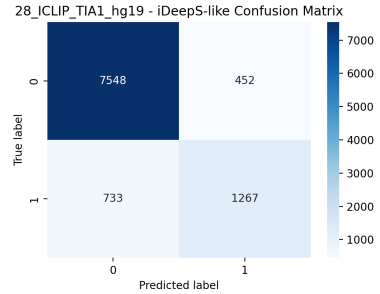
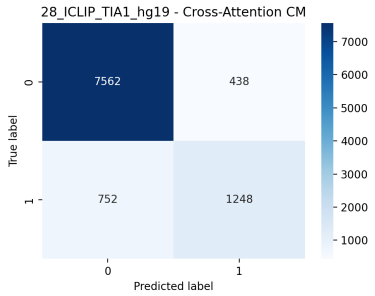
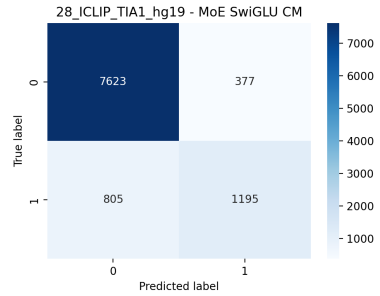


(c) Main iDeepS-like

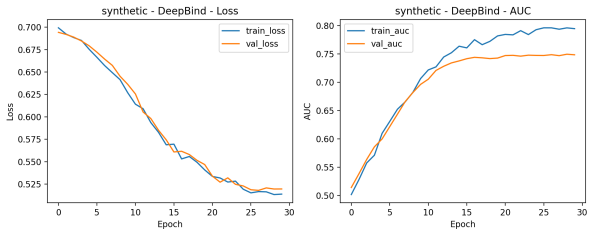


(d) Novel cross-attention

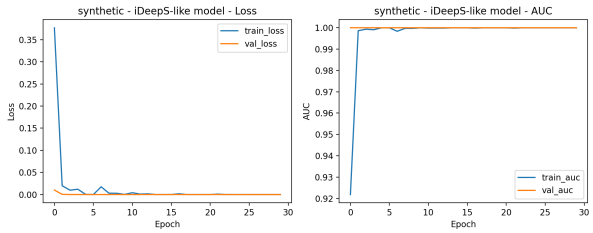


(e) Novel MoE-SwiGLU

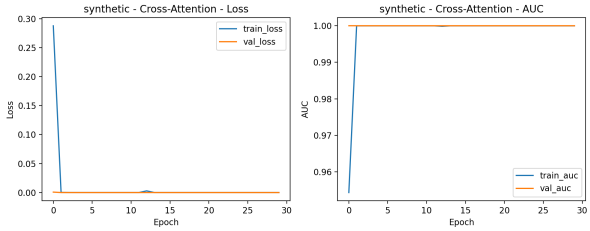
Figure 9: TIA1 confusion matrices for all five final models.



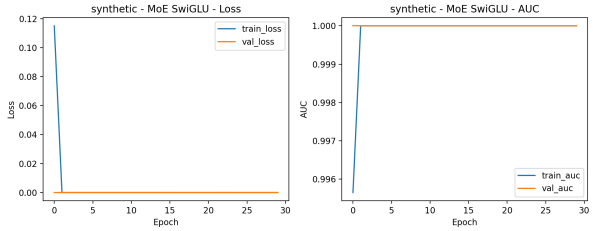
(a) DeepBind



(b) Main iDeepS-like

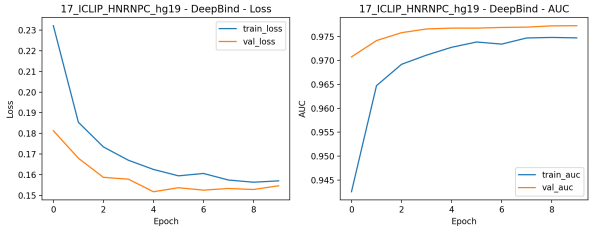


(c) Novel cross-attention

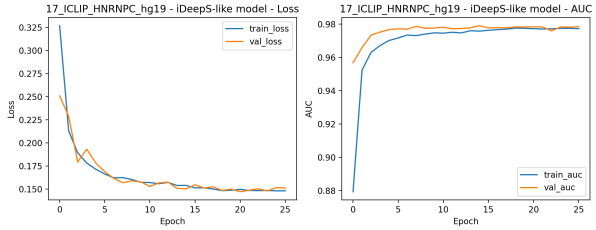


(d) Novel MoE-SwiGLU

Figure 10: Training histories on the synthetic dataset for the four deep models.



(a) DeepBind



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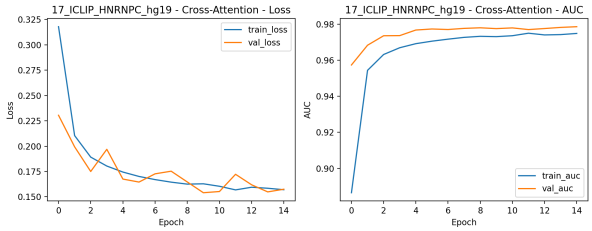
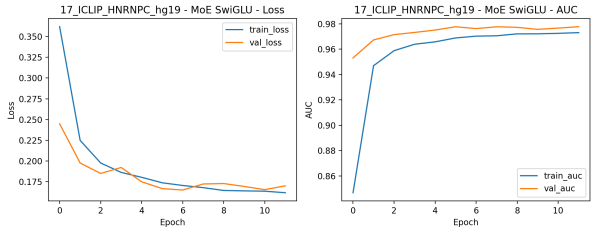
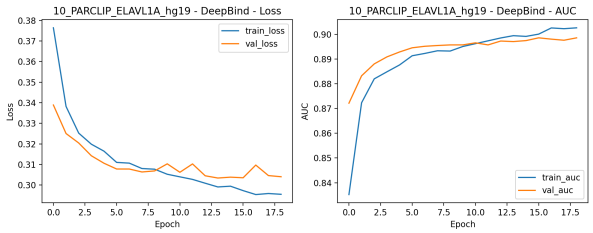


(c) Novel cross-attention

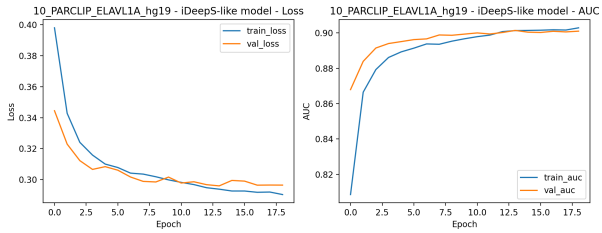


(d) Novel MoE-SwiGLU

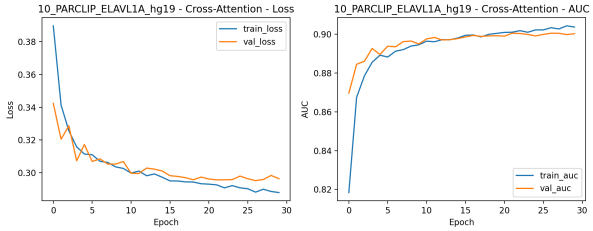
Figure 11: Training histories on HNRNPC for the four deep models.



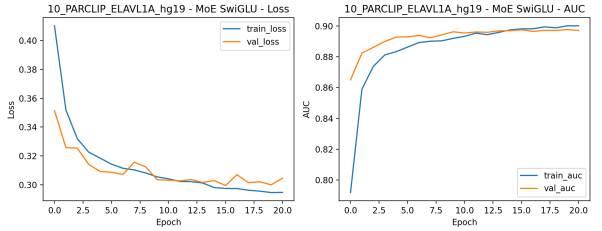
(a) DeepBind



(b) Main iDeepS-like

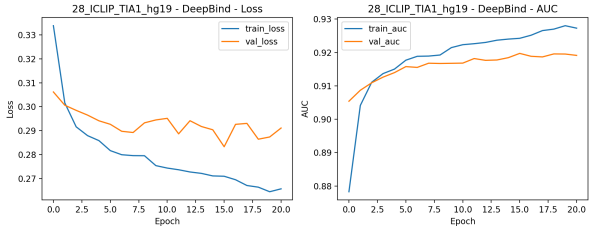


(c) Novel cross-attention

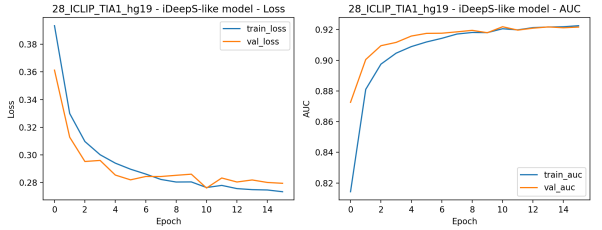


(d) Novel MoE-SwiGLU

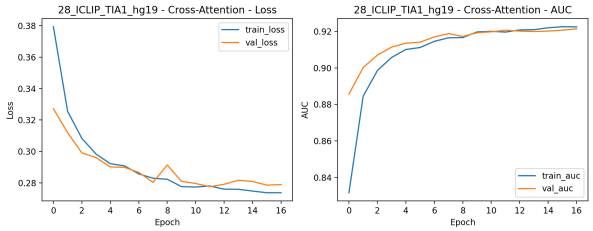
Figure 12: Training histories on ELAVL1A for the four deep models.



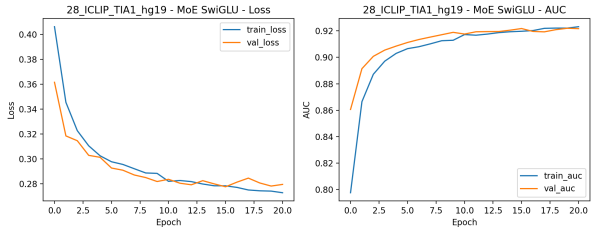
(a) DeepBind



(b) Main iDeepS-like



(c) Novel cross-attention



(d) Novel MoE-SwiGLU

Figure 13: Training histories on TIA1 for the four deep models.